Lecture 26: GARCH Models

1 Announcements

Sample final exam and final review sheet have been posted on Courseworks. The final review sheet contains all details about the exam. There will be a review session conducted by the TA Jose Sidaoui Gali on May 9, 2-3:30pm in 1127 MUDD. All details are also available in the final review sheet.

2 ARCH Models

How do we know if ARCH is necessary for the time series we are considering? Construct a residual plot, in which we plot $\epsilon_t := r_t - \hat{r}_{t-1}(1)$, where r_t follows an ARMA model. First, we visually check if the residual plot displays any cluster of volatility. Second, we can display the plot of squared residuals of the time series. If the volatility is constant, the variance of the error terms should be close to constant through time. If we observe that the variances of data are different over different periods of time, then it means that a better model is needed to capture the stochastic behavior of the volatility.

Consider the monthly log stock returns of Intel Corporation from January 1973 to December 2008 shown in Figure 1. It suggests no significant serial correlation (except for a minor one at lag 7). Moreover, it shows that the sample ACF of the absolute log returns (i.e., $|r_t|$) and the squared log returns r_t^2 are more significant. Combining these plots, we deduce that the log returns are *serially uncorrelated but dependent.* Volatility models attempt to capture such dependence in the return series.

The first model that provides a systematic framework for volatility modeling is the *Auto-Regressive conditional heteroscedastic model* (ARCH) model of Engle (1982). The basic idea of an $ARCH(1)$ model is that the shocks $\{a_t\}$ of asset returns are serially uncorrelated, but dependent, and the dependence of a_t can be described by a quadratic function of its lagged values. The model is specified by

$$a_t = \sigma_t \epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2,$$

where $\{\epsilon_t\}$ is a white noise sequence with zero mean and variance 1, $\alpha_0 > 0$ and $\alpha_1 \geq 0$ are constants.

In practice, ϵ_t is assumed to follow the standard normal or the Student's t distribution. Note that the variance at time t is connected to the value of the series at time $t-1$. A large past squared shock a_{t-1}^2 implies a larger conditional variance σ_t^2 for the innovation a_t. Consequently a_t tends to assume a large value in absolute value. This implies that, under the ARCH framework, a large shock tends to be followed by another large shock. *Conditional heteroskedasticity* is said to occur when the variance of the shock a_t, conditioned on the past shocks $a_1, \ldots, a_{t-1}$ changes dynamically

with t. In other words, σ_t^2 is not constant but dynamically changes over time as new shocks are realized. It is called *conditional heteroscedasticity* to emphasize the fact that it is the sequence of conditional variances that changes and not the unconditional variance.

Notice that $\{a_t\}$ is a serially uncorrelated zero-mean time series. Indeed,

$$\begin{aligned}
\mathbb{E}\left[a_t\right] &= \mathbb{E}\left[\mathbb{E}\left[a_t\middle|\mathcal{F}_{t-1}\right]\right] = \mathbb{E}\left[\sigma_t \mathbb{E}\left[\epsilon_t\middle|\mathcal{F}_{t-1}\right]\right] = 0 \\
Cov(a_t, a_{t-l}) &= \mathbb{E}[a_t a_{t-l}] = \mathbb{E}\left[\mathbb{E}[a_t a_{t-l}|\mathcal{F}_{t-1}]\right] \\
&= \mathbb{E}\left[\mathbb{E}[\sigma_t \epsilon_t \sigma_{t-l}\epsilon_{t-l}|\mathcal{F}_{t-1}]\right] = \mathbb{E}\left[\sigma_t\sigma_{t-l}\epsilon_{t-l}\mathbb{E}[\epsilon_t|\mathcal{F}_{t-1}]\right] = 0,
\end{aligned}$$

for any $l \geq 1$. We have used that ϵ_t is a white noise sequence, and that $\sigma_t, \sigma_{t-l}, \epsilon_{t-l}$ are all known at time $t-1$. Notice that

$$\begin{aligned}
Var(a_t) &:= \mathbb{E}\left[a_t^2\right] = \mathbb{E}\left[\left(\alpha_0 + \alpha_1 a_{t-1}^2\right)\epsilon_t^2\right] \\
&= \mathbb{E}\left[\left(\alpha_0 + \alpha_1 a_{t-1}^2\right)\mathbb{E}\left[\epsilon_t^2\middle|\mathcal{F}_{t-1}\right]\right] \\
&= \mathbb{E}\left[\alpha_0 + \alpha_1 a_{t-1}^2\right]
\end{aligned}$$

using that $Var(\epsilon_t) = 1$. If the time series a_t is weakly stationary, it follows immediately that

$$Var(a_t) = \frac{\alpha_0}{1-\alpha_1}.$$

Since the variance of a_t must be positive, we require $0 \leq \alpha_1 < 1$.

3 Leptokurtic Feature of ARCH Models

Consider

$$a_t = \sigma_t \epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2,$$

where $\{\epsilon_t\}$ is a white noise sequence with zero mean and variance 1. Let us analyze the *leptokurtic* feature of the ARCH time series. The leptokurtic feature of a frequency distribution is the extent to which the distribution has greater kurtosis than the normal distribution, i.e, it has fatter tails than the normal distribution. The more extreme points along the x-axis of the distribution are also called outliers.

Suppose that $\{a_t\}$ has stationary fourth order moment. Under the normality assumption of ϵ_t, and recalling that the fourth moment of a zero-mean and unit variance normal distribution is 3, we have that the conditional 4-th order moment is

$$\mathbb{E}\left[a_t^4\middle|\mathcal{F}_{t-1}\right] = \left(\alpha_0 + \alpha_1 a_{t-1}^2\right)^2 \mathbb{E}\left[\epsilon_t^4\middle|\mathcal{F}_{t-1}\right] = 3\left(\alpha_0 + \alpha_1 a_{t-1}^2\right)^2.$$

The unconditional 4-th order moment is

$$\begin{aligned}
\mathbb{E}\left[a_t^4\right] &= 3\left(\alpha_0^2 + 2\alpha_0\alpha_1\mathbb{E}\left[a_{t-1}^2\right] + \alpha_1^2\mathbb{E}\left[a_{t-1}^4\right]\right) \\
&= 3\left(\frac{\alpha_0^2(1+\alpha_1)}{1-\alpha_1} + \alpha_1^2\mathbb{E}\left[a_{t-1}^4\right]\right)
\end{aligned}$$

Because of stationarity of the 4-th order moment, i.e., $\mathbb{E}[a_t^4] = \mathbb{E}\left[a_{t-1}^4\right]$, we obtain

$$\mathbb{E}\left[a_t^4\right] = \frac{3\alpha_0^2(1+\alpha_1)}{(1-3\alpha_1^2)(1-\alpha_1)}.$$

This result has two important implications. First, since the fourth moment of a_t is positive, we have that α_1 must satisfy the condition $1 - 3\alpha_1^2 > 0$, that is, $0 \leq \alpha_1^2 < 1/3$. Second, the unconditional kurtosis is

$$\frac{\mathbb{E}[a_t^4]}{Var(a_t)^2} = \frac{3(1-\alpha_1^2)}{1-3\alpha_1^2} > 3$$

Hence, the kurtosis increases with α_1. Notice that the kurtosis of a standard univariate Gaussian is 3. Hence, the above inequality implies that the tail distribution of a_t is heavier than that of a normal distribution. In other words, the shock a_t of a Gaussian ARCH(1) model is more likely than a Gaussian white noise series to produce "outliers". This is in line with the empirical finding that outliers appear more often in asset returns that what implied by a Gaussian i.i.d sequence. These properties continue to hold for higher order ARCH models, but the formulas become more complicated.

3.1 ARCH(m) model

The direct generalization of the ARCH(1) model is the ARCH(m) model given by

$$a_t = \sigma_t \epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{m} \alpha_i a_{t-i}^2,$$

where $\{\epsilon_t\}$ is a white noise with variance 1. α_0 and α_i's are nonnegative constants. Under the assumption that $\{a_t\}$ is stationary, we have that

$$Var(a_t) = \frac{\alpha_0}{1 - \sum_{i=1}^{m} \alpha_i}.$$

For the variance to be positive, we require that $\sum_{i=1}^{m} \alpha_i < 1$.

3.2 Test for an ARCH Effect

A popular method to test for an ARCH effect is the Lagrange multiplier test proposed by Engle (1982). Let us assume that we have an $ARMA(p,q)$ model for the return series r_t. We build the *mean equation* $a_t = r_t - \mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right]$. Since ARCH suggests that a_t^2 is dependent of lags of itself, we can test the hypothesis of no ARCH by regressing a_t^2 on lags of itself, and test whether the parameters of this regression are statistically and economically significant. The test is equivalent to the usual F statistic for testing $\alpha_i = 0$, $i = 1, \ldots, m$, in the linear regression

$$a_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \alpha_2 a_{t-2}^2 + \ldots + \alpha_m a_{t-m}^2 + e_t, \qquad t = m+1, \ldots, T$$

where e_t denotes the error term, m is a prespecified positive integer, and $T - m$ is the sample size.[1] Specifically, the null hypothesis is $H_0 := \alpha_1 = \alpha_2 = \cdots = \alpha_m = 0$. Let $SSR_0 = \sum_{t=m+1}^{T} (a_t^2 - \bar{\omega})^2$,

[1] One way to choose m is to compare loglikelihood values for different choices of m.

where $\bar{\omega} = \frac{1}{T-m}\sum_{t=m+1}^{T} a_t^2$ is the sample mean of a_t^2, and $SSR_1 = \sum_{t=m+1}^{T} \hat{e}_t^2$, where $\hat{e}_t = a_t^2 - \widehat{a_t^2}$ is the least square residual of the previous linear regression. Then we have

$$F = \frac{(SSR_0 - SSR_1)/m}{SSR_1/(T-2m-1)},$$

follows a F distribution with degrees of freedom m and $T-2m-1$ under the null hypothesis (we have $T-m$ observations and we are estimating $m+1$ parameters so that we lose $m+1$ degrees of freedom) Morover, mF is asymptotically distributed as a chi-squared distribution with m degrees of freedom under the null hypothesis. The decision rule is to reject the null hypothesis if $mF > \chi^2_{m,1-\alpha}$, where $\chi^2_{m,1-\alpha}$ is the $100(1-\alpha)$th percentile of χ^2_m.

3.3 Forecast of ARCH model

At the forecast origin T, the 1-step ahead forecast of σ^2_{T+1} is

$$\hat{\sigma}^2_T(1) := \mathbb{E}\left[\sigma^2_{T+1}\middle|\mathcal{F}_T\right] = \alpha_0 + \sum_{i=1}^{m} \alpha_i a^2_{T+1-i}.$$

The 2-step ahead forecast is

$$\begin{aligned}
\hat{\sigma}^2_T(2) &:= \mathbb{E}\left[\sigma^2_{T+2}\middle|\mathcal{F}_T\right] \\
&= \alpha_0 + \alpha_1 \mathbb{E}\left[\epsilon^2_{T+1}\sigma^2_{T+1}|\mathcal{F}_T\right] + \alpha_2 \mathbb{E}\left[a^2_T|\mathcal{F}_T\right] + \ldots \alpha_m a^2_{T+2-m} \\
&= \alpha_0 + \alpha_1 \mathbb{E}\left[\epsilon^2_{T+1}\sigma^2_{T+1}|\mathcal{F}_T\right] + \alpha_2 a^2_T + \ldots \alpha_m a^2_{T+2-m} \\
&= \alpha_0 + \alpha_1 \mathbb{E}\left[\epsilon^2_{T+1}\right]\mathbb{E}\left[\sigma^2_{T+1}|\mathcal{F}_T\right] + \alpha_2 a^2_T + \ldots \alpha_m a^2_{T+2-m} \\
&= \alpha_0 + \alpha_1 \hat{\sigma}^2_T(1) + \alpha_2 a^2_T + \ldots \alpha_m a^2_{T+2-m}
\end{aligned}$$

The l-step ahead forecast is

$$\hat{\sigma}^2_T(l) := \mathbb{E}\left[\sigma^2_{T+l}\middle|\mathcal{F}_T\right] = \alpha_0 + \sum_{i=1}^{m} \alpha_i \hat{\sigma}^2_T(l-i),$$

where $\hat{\sigma}^2_T(j) := a^2_{T+j}$ for $j \leq 0$.

3.4 How to estimate the ARCH(m) model?

Several likelihood functions are commonly used in ARCH estimation, depending on the distributional assumptions of ϵ_t. Suppose $a_1, a_2, \ldots, a_T$ are observed. Given $a_{t-1}, a_{t-2}, \ldots, a_{t-m}$, we have that σ_t is known. Under the assumption that $\{\epsilon_t\}_{t\geq 0}$ is Gaussian distributed, the conditional likelihood is given by

$$L\left(a_{m+1:T}\middle|\alpha_0, \alpha_{1:m}, a_{1:m}\right) = \prod_{t=m+1}^{T} \frac{1}{\sqrt{2\pi\sigma^2_t}} e^{-\frac{a^2_t}{2\sigma^2_t}}$$

Hence, the conditional log-likelihood is given by

$$\begin{aligned}\log L\left(a_{m+1:T}\middle|\alpha_0,\alpha_{1:m},a_{1:m}\right) &= -\frac{T-m}{2}\log\left(2\pi\right)\\ &\quad -\frac{1}{2}\sum_{t=m+1}^{T}\log(\sigma_t^2)-\sum_{t=m+1}^{T}\frac{1}{2\sigma_t^2}a_t^2.\end{aligned}$$

We can then use a nonlinear optimization algorithm to find the best parameter estimates.

4 Pro and Cons of ARCH model

We have already discussed the pros of the ARCH model, which are the modeling of conditional heteroscedasticity, i.e. time varying variance that is able to capture clustering effects. However, the following are important cons of ARCH models.

- The model assumes that positive and negative shocks have the same effects on volatility, because the conditional variance depends on the square of the previous shocks. It practice, it is widely known that the price of a financial asset reacts differently to positive and negative shocks.
- The model is restrictive. To have finite fourth-order moment, we require that $\alpha_1^2 < 1/3$ in a ARCH(1). This constraint becomes complicated in higher order ARCH models. In practice, it limits the ability to capture excess kurtosis.
- It provides a mechanical way to describe the behavior of the conditional variance. But, it is silent on what causes such behavior to occur.

5 GARCH Model

The ARCH model is simple, but often requires many parameters to adequately describe the volatility process of an asset return. Bollerslev (1986) proposes a useful extension known as the generalized ARCH model, also referred to as the GARCH model. Let $a_t = r_t - \mathbb{E}\left[r_t\middle|\mathcal{F}_{t-1}\right]$, where r_t is a log-return series. Then the innovation a_t at time t follows a $GARCH(m,s)$ model if

$$a_t = \sigma_t\epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{m}\alpha_i a_{t-i}^2 + \sum_{j=1}^{s}\beta_j\sigma_{t-j}^2, \tag{1}$$

where $\{\epsilon_t\}_{t\geq 0}$ is a sequence of i.i.d. random variables with zero mean and variance 1, $\alpha_0 > 0$, $\alpha_i \geq 0$, and $\beta_j \geq 0$. As in the case of the ARCH model, it is often assumed that ϵ_t follows a standard normal or a generalized Student's t-distribution. If $s = 0$, the model reduces to a pure $ARCH(m)$ model. The α_j and β_j coefficients are referred to as ARCH and GARCH parameters, respectively. To understand properties of GARCH models, it is informative to use the following representation. Let $\eta_t = a_t^2 - \sigma_t^2$ so that $\sigma_t^2 = a_t^2 - \eta_t$. By plugging $\sigma_{t-j}^2 = a_{t-j}^2 - \eta_{t-j}$, $j = 1,\ldots,s$ into Eq. (1), we may rewrite the GARCH model as

$$a_t^2 - \eta_t = \alpha_0 + \sum_{i=1}^{m}\alpha_i a_{t-i}^2 + \sum_{j=1}^{s}\beta_j a_{t-j}^2 - \sum_{j=1}^{s}\beta_j\eta_{t-j},$$

or, equivalently, as

$$a_t^2 = \alpha_0 + \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{s} \beta_j \eta_{t-j},$$

where it is understood that $\alpha_i = 0$ for $i > m$ and $\beta_j = 0$ for $j > s$. It can be easily checked that $\{\eta_t\}_{t\geq 0}$ has zero mean and it is a serially uncorrelated time series, i.e., $cov(\eta_t, \eta_{t-j}) = 0$ for $j \geq 1$ (Notice, however, that $\{\eta_t\}$ is not a i.i.d. sequence in general). In light of this, a GARCH model may be viewed as an application of the ARMA model to the squared series a_t^2. Using the expression for the unconditional mean of an ARMA model, we obtain that

$$\mathbb{E}\left[a_t^2\right] = \frac{\alpha_0}{1 - \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i)},$$

provided that the denominator is positive. Hence, we need to impose the restriction $\sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1$ for a $GARCH(m, s)$ model to be well defined.

The strengths and weaknesses of GARCH models can be seen easily by focusing on the simplest GARCH(1,1) model where

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad 0 \leq \alpha_1, \beta_1 \leq 1, (\alpha_1 + \beta_1) < 1. \tag{2}$$

First, a large a_{t-1}^2 or σ_{t-1}^2 gives rise to a large σ_t^2. This means that a large a_{t-1}^2 tends to be followed by another large a_t^2, generating again the well known behavior of volatility clustering in financial time series. Second, it can be shown (as in the case of the ARCH model) that the kurtosis of the distribution of a_t is larger than 3, i.e.,

$$\frac{\mathbb{E}\left[a_t^4\right]}{\left(\mathbb{E}\left[a_t^2\right]\right)^2} = \frac{3\left(1 - (\alpha_1 + \beta_1)^2\right)}{1 - (\alpha_1 + \beta_1)^2 - 2\alpha_1^2} > 3.$$

Consequently, the tail distribution of a GARCH(1,1) process is heavier than that of a normal distribution. Third, the model provides a simple parametric function to describe the volatility evolution.

Forecasts of a GARCH model can be obtained using methods similar to those of an ARMA model. Consider the GARCH(1,1) model and assume the forecast origin is T. Then

$$\hat{\sigma}_T^2(1) = \alpha_0 + \alpha_1 a_T^2 + \beta_1 \sigma_T^2$$

For multistep-ahead forecasts, we use $a_t^2 = \sigma_t^2 \epsilon_t^2$, and rewrite the volatility equation in (2) as

$$\sigma_{t+1}^2 = \alpha_0 + (\alpha_1 + \beta_1)\sigma_t^2 + \alpha_1 \sigma_t^2 (\epsilon_t^2 - 1).$$

When $t = T + 1$, the equation becomes

$$\sigma_{T+2}^2 = \alpha_0 + (\alpha_1 + \beta_1)\sigma_{T+1}^2 + \alpha_1 \sigma_{T+1}^2 (\epsilon_{T+1}^2 - 1).$$

This leads to

$$\hat{\sigma}_T^2(2) = \alpha_0 + (\alpha_1 + \beta_1)\hat{\sigma}_T^2(1).$$

In general, we have

$$\hat{\sigma}_T^2(l) = \alpha_0 + (\alpha_1 + \beta_1)\hat{\sigma}_T^2(l-1), \qquad l > 1. \tag{3}$$

This result is exactly the same as that of an ARMA(1,1) model with AR polynomial $1-(\alpha_1+\beta_1)B$, where B is the back-shift operator. By repeated substitutions in Eq. (3), we obtain that the l-step ahead forecast is

$$\hat{\sigma}_T^2(l) = \frac{\alpha_0\left[1-(\alpha_1+\beta_1)^{l-1}\right]}{1-\alpha_1-\beta_1} + (\alpha_1+\beta_1)^{l-1}\hat{\sigma}_T^2(1).$$

As $l \to \infty$, $\hat{\sigma}_T^2(l) \to \frac{\alpha_0}{1-\alpha_1-\beta_1}$. Hence, the multi-steps ahead volatility forecast of a GARCH(1,1) model converges to the unconditional variance of a_t as the forecast horizon increases to infinity, provided that $Var(a_t)$ exists.

The conditional maximum likelihood method continues to apply to estimate the model. There is an enormous literature on GARCH models: see, for instance, Bollerslev, Chou and Kroner (1992), Bollerslev, Engle and Nelson (1994).

The model suffers from the same weaknesses as ARCH. For instance, it responds equally to positive and negative shocks. Studies of high-frequency financial time series suggest that the tail behavior of GARCH models remains too short even if we use standardized Student t's distribution for the innovation terms.

Remark 5.1. *It can be shown that a $GARCH(m,s) = ARCH(\infty)$. This means that the conditional variance depends on an infinite number of past squared innovations. Hence, the GARCH is more parsimonious than an ARCH model, because it uses just a few parameters to achieve what the ARCH model would do with an infinite number of parameters. In other words, the conditional variance may have a long memory with low values of m and s. This argument is very similar to how an ARMA model is more parsimonious than an AR or MA model, because we would need a $MA(\infty)$ or $AR(\infty)$ representation to achieve what an ARMA model does with a few parameters. In general, with low values of m and s, one can generate complex features, whereas you would need a higher value of m to generate the same features using an ARCH model. This means that you would be able to generate, say, an ACF of squared returns for the GARCH model with a richer pattern than if using an ARCH model with a similar number of parameters. Lower order GARCH models are used in most applications, say GARCH(1,1), GARCH(2,1) and GARCH(1,2) models.*

6 Relation to Stochastic Volatility Models

A popular stochastic volatility model is

$$a_t = \sigma_t \epsilon_t, \qquad \log(\sigma_t^2) = \alpha_0 + \sum_{i=1}^{m} \alpha_i \log(\sigma_{t-i}^2) + \nu_t,$$

where $\{\epsilon_t\}$ is a white noise sequence, α_0, α_i's are non-negative constants, $\{\nu_t\}$ is a white noise sequence with variance σ_ν^2, $\{\epsilon_t\}$ and $\{\nu_t\}$ can be correlated. Indeed, stock asset returns and volatility are negatively correlated. Such a negative correlation goes under the name of leverage effect, and we have discussed this in the context of ARCH models. We refer to Melino and Turnbull (1990), Harvey, Ruiz, and Shephard (1994) and Jacquier, Polson, and Rossi (1994), for a study of these models.

There is an important difference between ARCH and stochastic volatility models. In ARCH, the conditional variance σ_t^2 is a deterministic function of past a_t^2's. In stochastic volatility models, the conditional variance involves a new innovation term ν_t.

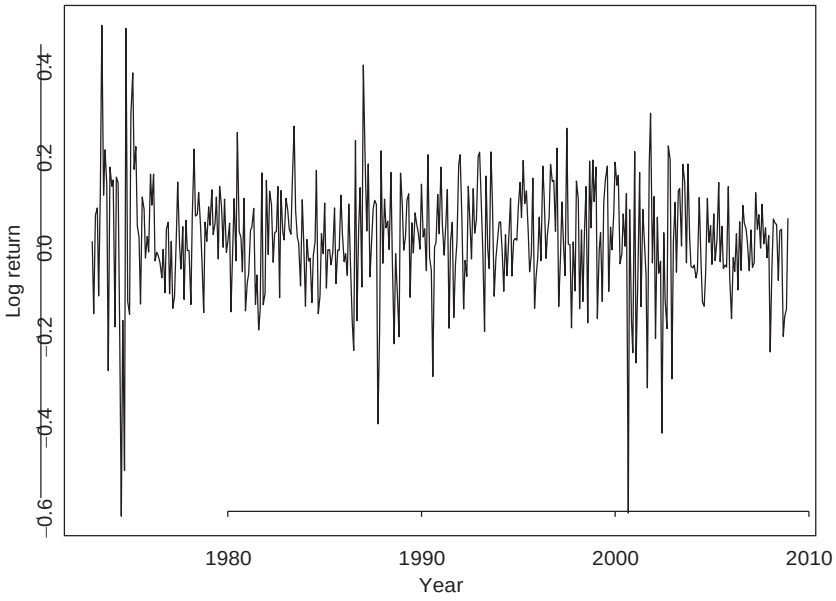


Figure 3.1 Time plot of monthly log returns of Intel stock from January 1973 to December 2008.

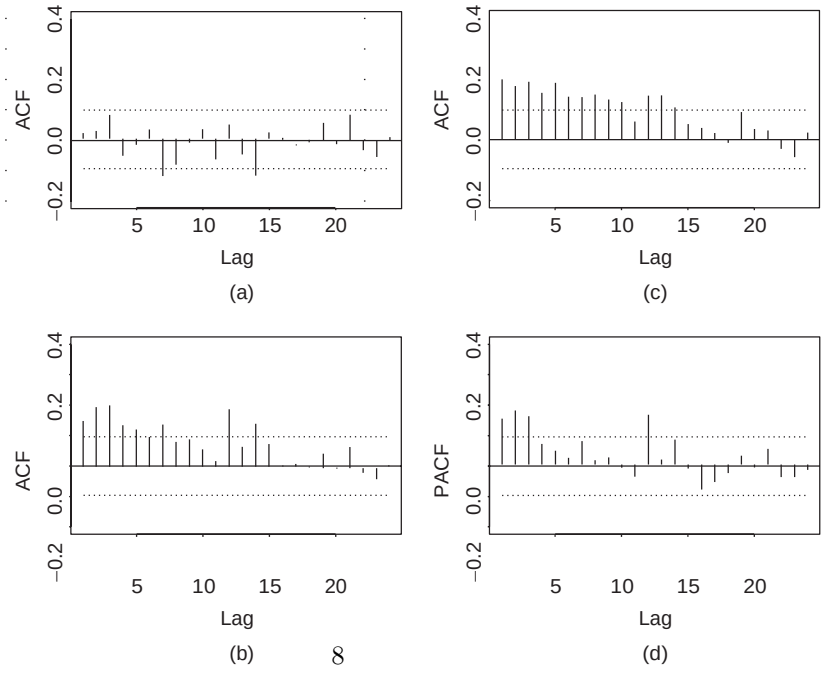


Figure 3.2 Sample ACF and PACF of various functions of monthly log stock returns of Intel Corporation from January 1973 to December 2008: (a) ACF of the log returns, (b) ACF of the squared log returns, (c) ACF of the absolute log returns, and (d) PACF of the squared log returns.